Final Year Project

Timar

A Digital Platform for Mangrove Restoration on the Coast of Karachi

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1. Introduction

Timar is a web platform built for the people who are keeping Karachi's mangrove forests alive. The word **timar** is what these salt water forests are called locally. They grow where the city meets the sea, along Korangi Creek, Phitti Creek and Sandspit, and their roots hold the coastline in place. Most people living in Karachi have never visited them and many do not know they are there.

Saltwater Marina is a community led organisation working on mangrove restoration along this coastal belt. They organise plantation drives, take volunteers out to the creeks, and plant seedlings by hand. What they do not have is any system to record what was planted, where it was planted, and whether it survived. That record lives in photographs on personal phones, in WhatsApp groups, and in the memory of whoever was present that day.

Timar gives that work a digital home. The platform lets anyone book a trip to a plantation site, sponsor a seedling and follow what happens to it, read about why these forests matter, and report illegal cutting when they see it. For Saltwater Marina it becomes an operations system: a map of every plantation plot, survival counts recorded over time, volunteer records, and reports they can hand to a donor or a company. Underneath all of this sits a set of machine learning models that read field photographs, count seedlings, suggest where to plant next, and watch satellite imagery for signs of cutting.

2. Objective

The goal of this project is to replace guesswork with records. Restoration work in Karachi is being done sincerely, but it is not being measured, and what is not measured cannot be improved, funded or defended. Timar is built to measure it.

2.1 Specific Objectives

These are the outcomes the project will be judged on, not a list of screens. The modules that deliver them are described in Section 6.

- O1: Give the public a clear and honest explanation of what mangroves do for Karachi, and make that information easy to find and share.
- O2: Record every plantation site as a mapped plot with its boundary, species, planting date and responsible person, so that no planting event is lost.
- O3: Measure survival instead of assuming it, by running scheduled field surveys at fixed intervals and storing the counts against the plot.
- O4: Let a person or a company who funded seedlings see exactly what happened to them, including coordinates, a live survival counter and follow up photographs.
- O5: Make it simple to become a volunteer, to be given field work, and to see a personal record of contribution.

- O6: Make losses visible by letting anyone file a located complaint about cutting or encroachment, and by flagging suspected loss from satellite imagery.
- O7: Reduce the manual effort of field monitoring using machine learning that assesses plant health from photographs, counts seedlings, scores candidate planting sites and detects change over time.
- O8: Give Saltwater Marina a single administrative portal to run trips, volunteers, plots, content and reporting, with an audit trail behind every change.
- O9: Provide a community space where people interested in the coast and the climate can post, discuss and share their own field photographs.

3. Problem Description

It would be easy to say that Pakistan is losing its mangroves, but that would not be accurate. Pakistan is one of the few countries in the world where mangrove cover has grown, from roughly **48,000 hectares** in **1986** to around **144,000 hectares** in **2020**, largely because of sustained plantation work. The problem is not the absence of planting. The problem is everything that is missing around it.

3.1 What is actually going wrong

First, most of what is planted does not live. Initial survival rates in Pakistan sit at around **40 to 45 percent**. Plots are replanted to make up the difference, but there is no record of which plots failed, in which season, with which species, so the same mistakes are repeated.

Second, losses go unrecorded. Karachi's coastline lost roughly **200 hectares** of mangrove between **2010** and **2022** to housing schemes, land reclamation and commercial cutting near **Port Qasim**, and reporting on the subject notes that there is very little data tracking these losses as they happen. Where the barrier has already gone, the cost is visible. Around **2.2 million acres** of delta farmland has turned unproductive from sea water, and more than **1.2 million** people have been displaced from the coastal belt.

Third, nobody can prove impact. A company that pays for ten thousand seedlings has no way to check what survived, which keeps serious climate funding and carbon finance out of reach for small community organisations, even though mangroves absorb carbon two to four times faster than mature tropical forests.

Fourth, willing people stay disconnected. Volunteers, plantation drives, awareness efforts and complaints all pass through informal chat groups and then disappear. There is no front door for someone who simply wants to help.

Fifth, and underneath all of it, awareness is very low. When a cyclone warning is issued for Karachi, something has to absorb the water before it reaches the city. In **June 2023**, when **Cyclone Biparjoy** approached, the Sindh coastal belt was largely spared, and the expanded

mangrove forest is credited with much of that protection. Very few residents of this city could name the forest that did it.

3.2 Problem Statement

Karachi's mangrove forests protect the city, yet most citizens do not know they exist, most planted seedlings die untracked, and the organisations restoring them have no digital system to record survival, prove impact or involve the public.

4. Target Industry

The primary users are environmental non profit organisations working on coastal restoration, with Saltwater Marina as the partner organisation for this project. Beyond that the platform serves four other groups.

Group	What they use it for
Citizens and students	Learning what mangroves do, booking a trip, sponsoring a seedling, reporting cutting
Volunteers	Signing up, receiving field tasks, recording surveys, tracking their own contribution
Companies	Sponsoring plantation, receiving verifiable impact reports for climate and social responsibility commitments
Government and researchers	A located record of plantation, survival and reported losses along the Karachi coast

In industry terms the project sits across environmental technology, climate and carbon reporting, eco tourism, civic technology and geographic information systems.

5. Methodology

5.1 Development Approach

We will work in two week iterations with a working build at the end of each one. Saltwater Marina will be shown the build at the end of every iteration, because they are the people who will actually use it and their field reality will correct our assumptions faster than any specification document. Field data collection starts early, in the first semester, since the machine learning work depends entirely on photographs that do not exist yet.

5.2 System Architecture

The platform has four layers. The client is an installable progressive web application. A backend service handles accounts, roles, bookings, plots, surveys and reporting. A separate Python service hosts the machine learning models and the satellite processing, because that is where the libraries live. Data sits in a spatial database, since almost everything in this project is a location.

5.3 Where AI and Machine Learning Are Applied

Five learned components are planned. Four are core and one depends on data the platform itself will generate.

	Component	What it does	Approach	Priority
M1	Plant health classifier	Labels a seedling or plot photograph as healthy, stressed or dead	Transfer learning on a convolutional network	Core
M2	Seedling counter	Counts visible seedlings in a field photograph so survey counts are measured rather than estimated	Object detection with density estimation	Core
M3	Site suitability model	Scores a candidate planting location using satellite and tidal features	Gradient boosted model over spatial features	Core
M4	Change detection	Compares monthly satellite imagery to flag drops in green cover that may indicate cutting	Vegetation index differencing with a classifier on the change map	Core
M5	Survival risk prediction	Predicts which plots are likely to fail, from species, site, season and past survey results	Tabular model trained on the platform's own survey history	Stretch

M5 is listed honestly as a stretch goal. It only becomes possible because Timar is collecting survival data that does not currently exist anywhere, which is also what makes it the most interesting part of the project academically.

5.4 Data and Sources

Field photographs will be collected with Saltwater Marina during plantation drives and survey rounds, then labelled by us with their guidance. This is the main dataset for M1 and M2 and we expect it to be small, so transfer learning and image augmentation are planned from the start rather than added later.

Satellite imagery for M3 and M4 will come from the European Copernicus Sentinel-2 programme, which is openly available at ten metre resolution with a revisit of a few days. Existing mangrove extent maps such as Global Mangrove Watch will be used as reference layers. The licence and usage conditions of every external dataset will be read at source and recorded in the SRS before the data is used, and no dataset will be used on the basis of an assumption about its licence.

5.5 Evaluation

Each model has a number attached to it and a simple baseline to beat. M1 is measured on classification accuracy and F1 across its three classes. M2 is measured on counting error against hand counted photographs. M3 is measured on how closely its ranking of candidate sites agrees with the ranking given independently by the Saltwater Marina field team. M4 is measured on precision and recall against manually checked satellite tiles. If a model cannot beat its simple baseline, such as a plain vegetation index threshold for M4, it will not be kept.

5.6 Honesty and Limits

Two limits are stated plainly in the system itself. A photograph can show visible stress but it cannot certify that a plant will live, so the health output is presented as an assessment with a confidence value and not as a guarantee. Secondly, Timar does not issue carbon credits. Real credits are issued by accredited standards bodies after independent verification. What Timar produces is the evidence record that such a verification would need, and the company facing reports describe it in exactly those words.

6. Project Scope

6.1 Functional Modules

Thirteen modules in total, FM1 to FM13. All of them are part of the committed scope.

	Module	Description
FM1	Accounts and roles	Registration, login and role based access for visitors, volunteers, companies, field coordinators and administrators
FM2	Trip booking	Browse and book plantation drives and eco tourism trips, with capacity limits, attendance marking and a safety briefing acknowledgement
FM3	Sponsor a seedling	Buy one or more seedlings, then follow them through coordinates, a live survival counter and the option to request a follow up photograph
FM4	Plot registry and map	Every plantation site recorded as a mapped boundary with species, planting date, count and responsible person
FM5	Survival surveys	Scheduled field survey rounds with counts, notes and a photograph timeline for each plot taken from the same position over months
FM6	Awareness content	Articles, a species guide for the mangroves of the Indus Delta, and shareable explainers
FM7	Volunteer management	Volunteer applications, approval, task assignment by area, and a personal record of hours and plots worked
FM8	Complaint assistant	A located, photographed complaint about cutting or encroachment, compiled into a formal document, sent onward by the organisation and tracked by status, shown together on a public threat map
FM9	Company portfolio	A sponsorship page per company with its plots, survival figures, estimated carbon and downloadable impact reports
FM10	Public impact dashboard	Open to everyone without login: seedlings planted, survival percentage, area covered and plots verified
FM11	Admin portal	Saltwater Marina's own console for trips, volunteers, plots, content, media requests, reports and an audit log
FM12	Community feed	Posts, photographs and discussion between people interested in the coast and the climate
FM13	Media requests	A queue for interview and media requests to the Saltwater Marina team, with status and scheduling

6.2 Out of Scope

- Issuing or trading carbon credits. The platform produces evidence.
- Direct automated filing into any government system. Complaints are compiled and sent by the organisation, and their status is updated manually.
- Drone or satellite image capture of our own. We use openly available imagery.

- Payment gateway integration beyond a single simple provider, and no handling of card data on our servers.

7. Feasibility Study

7.1 Risks and How We Handle Them

Risk	How we reduce it
Not enough labelled field photographs for M1 and M2	Start collection in the first semester, use transfer learning and augmentation, and fall back to a smaller two class health model if needed
Field access is limited by tides and weather	Plan survey rounds around the tide calendar and treat every field visit as a scheduled task in the system
The complaint channel depends on a government department responding	Design the feature as assisted filing with manual status updates, so it is useful even with no response
Poor GPS accuracy inside the creeks	Record accuracy with every point, require the point to fall inside the plot boundary, and allow manual correction by a coordinator
Scope is wide for a two member team	Thirteen core modules with the lighter ones scheduled last, and one stretch AI component that can be dropped without affecting the rest

7.2 Resources Required

- Two development laptops, which we already have.
- Free cloud GPU hours on Colab or Kaggle for model training. We estimate under twenty GPU hours in total.
- Open satellite imagery from Copernicus, at no cost.
- A free tier cloud host and database for the development deployment.
- Field time with Saltwater Marina, and mobile phone cameras for data collection.

8. Solution Application Areas

The same platform, with a different species list and a different partner, applies to any community led restoration effort. Inland tree plantation drives, coral and seagrass restoration, urban forest projects and watershed rehabilitation all share the same underlying problem of planting without proof. Beyond restoration, the plot and survey model fits agricultural extension work and any field programme where someone must visit a location, count something and prove they were there.

9. Tools and Technology

The choices below are our current plan and may change as the project proceeds.

Layer	Planned tools
Frontend	React with Tailwind CSS
Maps	MapLibre or Leaflet for the plot and threat maps

Layer	Planned tools
Backend	Node.js with Express, token based authentication and role based access control
AI services	Python with FastAPI, PyTorch for the image models and scikit-learn for the tabular ones
Geospatial	PostgreSQL with the PostGIS extension, Sentinel-2 imagery from Copernicus, processed with rasterio

10. Expertise of the Team Members

Both team members work full stack. We are deliberately not dividing the project into a frontend person and a backend person, because we want both of us to understand every layer of the system by the end of it. Work will be shared task by task, and whoever takes a task owns it from the database up to the screen.

11. Milestones

11.1 FYP-I, First Semester

The target for the first semester is a working platform that records reality, plus the first two models.

- MS1: Requirements gathered with Saltwater Marina, SRS written.
- MS2: Database and spatial schema designed.
- MS3: Accounts, roles and the public site live.
- MS4: Plot registry and map working, with real plots entered.
- MS5: Survival survey rounds and the photograph timeline working.
- MS6: Field photograph collection and labelling underway.
- MS7: M1 plant health classifier trained and connected.
- MS8: M2 seedling counter trained and connected.
- MS9: Prototype demonstration and FYP-I documentation.

11.2 FYP-II, Second Semester

The second semester completes the public and organisational features, adds the satellite work, and evaluates everything.

- MS10: Trip booking and sponsor a seedling with live tracking.
- MS11: Volunteer management and task assignment.
- MS12: M3 site suitability model and the recommendation view for the field team.
- MS13: M4 satellite change detection with monthly alerts.
- MS14: Complaint assistant and the public threat map.
- MS15: Company portfolio, impact reports and the public impact dashboard.

- MS16: Admin portal completed with the audit log, and the community feed delivered.
- MS17: Evaluation of all models against their targets, system and user testing.
- MS18: Media requests module, then final documentation and defence.

12. Project Schedule

Each semester is sixteen weeks. Phases overlap on purpose, so that every model can be tested inside a working application instead of on its own.

12.1 FYP-I, Fall Semester

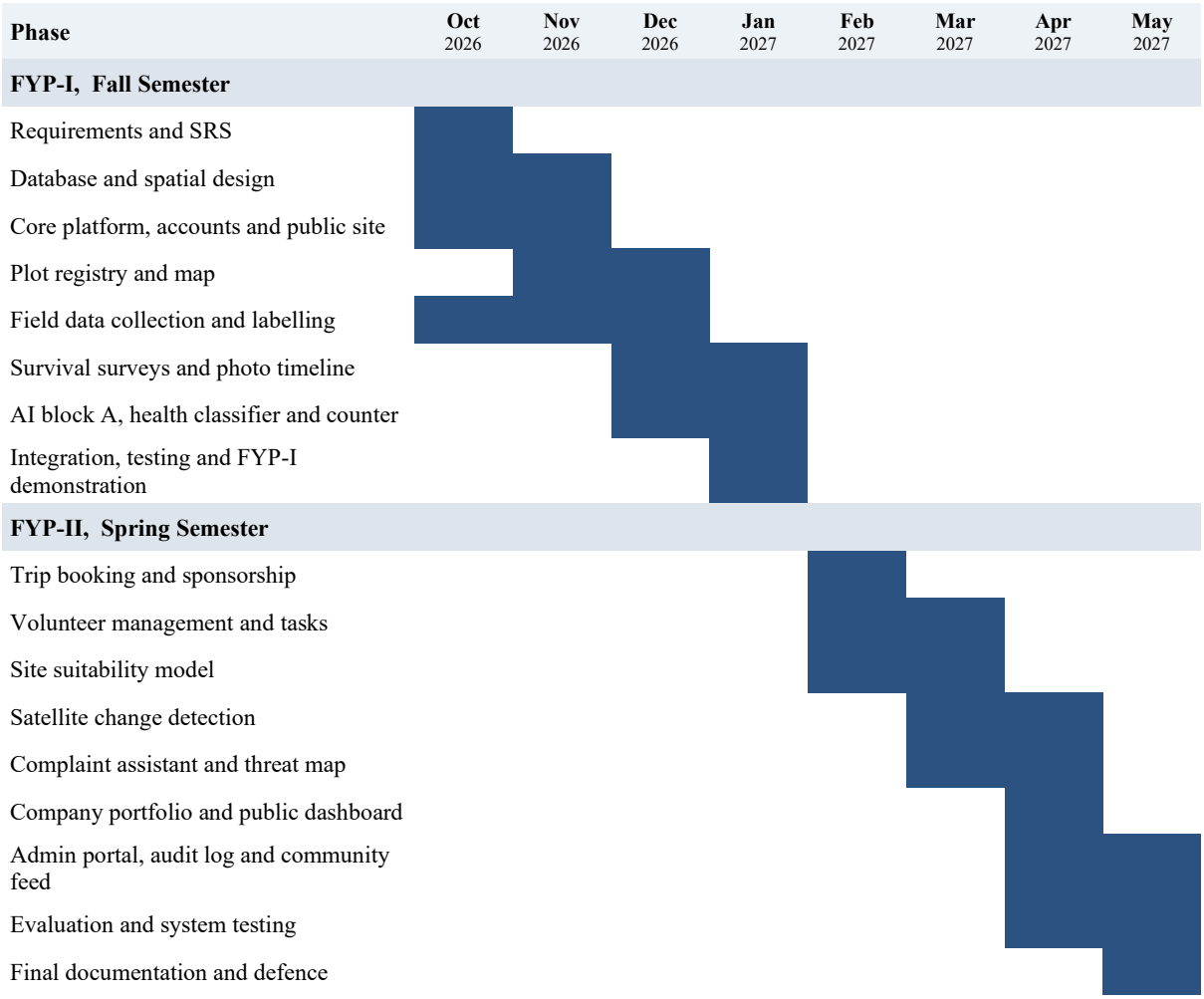
	Task	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1	Requirements and SRS	■	■	■	■												
2	Database and spatial design			■	■	■	■										
3	Accounts, roles, public site				■	■	■	■	■								
4	Plot registry and map						■	■	■	■	■						
5	Field data collection and labelling				■	■	■	■	■	■	■	■	■				
6	Survival surveys and photo timeline									■	■	■	■	■			
7	M1 health classifier										■	■	■	■	■		
8	M2 seedling counter												■	■	■	■	
9	Integration, testing, demo													■	■	■	■

12.2 FYP-II, Spring Semester

	Task	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
10	Trip booking and sponsorship	■	■	■	■	■											
11	Volunteer management			■	■	■	■	■									
12	M3 site suitability model		■	■	■	■	■	■	■								
13	M4 satellite change detection					■	■	■	■	■	■	■					
14	Complaint assistant and threat map							■	■	■	■	■					
15	Company portfolio and reports								■	■	■	■	■				
16	Admin portal and audit log								■	■	■	■	■	■	■		
17	Evaluation and testing											■	■	■	■	■	
18	Final documentation and defence													■	■	■	■

12.3 Month Wise Schedule

The same plan seen by calendar month rather than by week. October 2026 to January 2027 is the first semester and February 2027 to May 2027 is the second, which makes it easier to see how the two halves of the project join up.



13. Work Breakdown Structure

13.1 FYP-I

- **1.0 Planning:**

  1.1 requirements with Saltwater Marina,

  1.2 SRS,

  1.3 database and spatial schema

- **2.0 Data:**

  2.1 field photograph collection,

  2.2 labelling,

  2.3 satellite data access and licence check

- **3.0 Core platform:**

  3.1 accounts and roles,

  3.2 public site and content,

  3.3 plot registry,

  3.4 map views
- **4.0 Monitoring:**

  4.1 survey rounds,

  4.2 photograph timeline,

  4.3 verification rules
- **5.0 AI block A:**

  5.1 train M1,

  5.2 train M2,

  5.3 connect models to the platform
- **6.0 Review:**

  6.1 prototype testing,

  6.2 FYP-I documentation and demonstration

13.2 FYP-II

- **7.0 Public features:**

  7.1 trip booking,

  7.2 sponsor a seedling and tracking,

  7.3 volunteer management
- **8.0 AI block B:**

  8.1 build M3 site suitability,

  8.2 build M4 change detection,

  8.3 attempt M5 survival risk
- **9.0 Accountability:**

  9.1 complaint assistant,

  9.2 threat map,

  9.3 company portfolio and reports,

  9.4 public dashboard
- **10.0 Organisation tools:**

  10.1 admin portal,

10.2 audit log,

10.3 community feed and media requests

- **11.0 Closure:**

  11.1 model evaluation,

  11.2 system and user testing,

  11.3 final documentation and defence

14. References

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